

AMOLAK SINGH

Senior Mobile Architect & Tech Lead

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PROFESSIONAL SUMMARY

Highly analytical **Senior Mobile Architect & Tech Lead** with 9 years of professional expertise architecting, scaling, and optimizing mission-critical native Android and cross-platform mobile ecosystems. Specialized in reactive multi-module codebases, Clean Architecture, and predictable state management paradigms (MVI/MVVM). Proven track record leading cross-functional engineering teams, accelerating development velocity, slashing app cold-start latencies by 30%, and reducing crash rates by 40%. Expert at leveraging AI-assisted development workflows (Claude, Cursor) as a full-stack force multiplier to streamline backend integration pipelines and rapid prototyping.

CORE COMPETENCIES

• Native Android:	Kotlin, Java, Jetpack Compose, Coroutines, Flow, StateFlow, Channels, Jetpack Navigation, LiveData
• Architecture & Patterns:	Multi-Module Architecture, Clean Architecture, MVI, MVVM, Repository Pattern, Offline-First Hydration
• Cross-Platform & Backend:	Kotlin Multiplatform (KMP), Compose Multiplatform (CMP), Flutter, React Native, Ktor Client, Node.js primitives
• Data & Networking:	Retrofit, Room DB, SQLite, DataStore, Firebase Suite (Auth, Firestore, Cloud Messaging, Crashlytics)
• DI & Quality Tooling:	Dagger Hilt, Koin, JUnit 4/5, MockK, Turbine, GitHub Actions, CI/CD, Gradle Optimization
• AI-Assisted Development:	Claude AI, Cursor, Prompt Engineering, Rapid Prototype Full-Stack Architecture Optimization

PROFESSIONAL EXPERIENCE

Senior Mobile Architect & Tech Lead <i>CodeBrew Labs (incl. BlockTech Brew Venture)</i>	Feb 2020 – Present <i>Chandigarh, India</i>
• Ecosystem Leadership: Spearheaded mobile architecture and engineering governance across 10+ enterprise Android and cross-platform applications and internal ventures (including BlockTech Brew), driving a 35% growth in global consumer downloads. • System Modularization: Standardized reactive Multi-Module layered architecture (Core/Data/Domain/UI) across active business lines, maximizing code reusability and sharing business logic up to 90% across platforms. • Performance Engineering: Reduced client cold-start latencies by 30% through strict memory profiling, leak mitigation, dependency injection optimization (Dagger Hilt/Koin), and custom rendering metrics. • Engineering Governance: Chaired a 6-member cross-functional mobile team; instituted structured PR code-review governance, automated CI/CD workflows via GitHub Actions, and streamlined delivery pipelines. • Quality Standards: Reduced code defect counts by 40% via granular Firebase Crashlytics monitoring, automated regression pipelines, and automated end-to-end telemetry testing.	

Android Application Engineer

Crinoid Technologies

Aug 2017 – Jan 2020

Chandigarh, India

- **Product Engineering:** Shipped 5 major full-scale native Android solutions to production markets (totaling 500K+ cumulative installs) while boosting sprint planning accuracy and efficiency by 30%.
- **Network Caching & Logic:** Cut active network API response latency benchmarks by 25% through offline-first Room DB caching structures, state sync interceptors, and Retrofit optimization.
- **App Reliability:** Enhanced app stability and user retention by implementing structured unit testing protocols, RxAndroid/LiveData state management, and rapid bug mitigation workflows.

FEATURED HIGH-IMPACT PROJECTS

Multi-Platform Live Social Streaming & Media Engine

Technologies: Jetpack Compose, Multi-Module, MVI, RTMP/RTMPS, OpenGL

- Architected an enterprise high-concurrency streaming engine using RTMP/RTMPS protocols, enabling simultaneous multi-platform broadcasting. Integrated custom dynamic watermark/OpenGL video overlays directly onto live feeds and engineered dynamic map clustering algorithms to enhance user discovery.

Astrology Interactive Platform (User + Consultant Ecosystem)

Technologies: Jetpack Compose, Kotlin, Clean Architecture, WebSockets, Coroutines

- Designed an asymmetric dual-app streaming architecture facilitating instant broadcast synchronization, stateful session recovery handlers, and highly optimized multi-layered atomic transitions across high-frequency disconnects.

Commbitz International eSIM Marketplace

Technologies: Kotlin, Multi-Module, Domain-Driven UI, Retrofit, Room DB

- Engineered a global telecommunication data marketplace covering 100+ countries with localized multi-currency purchasing funnels, driving an immediate 15% upward trend in direct platform conversions.

Coown – Fractional Property Investment Platform

Technologies: Compose Multiplatform (CMP), Ktor, MVI, StateFlow

- Built multi-layered transactional pipelines supporting real-time data visualizers, secure asset ownership transfers, and atomic local-to-cloud transactional synchronizations using Ktor client and Compose Multiplatform.

EDUCATION & CERTIFICATIONS

- **Master of Computer Science (MCS)** – Lovely Professional University (GPA: 7.39 / 10)
- **Bachelor of Computer Applications (BCA)** – Arjan Dass College
- **Certifications:** Advanced Android Development (CBitss Institute) | Python & Scripting Systems (CBitss)